

## Lab program-19:

19. Design a C program to implement process synchronization using mutex locks.

Code:

```
#include <stdio.h>
#include <windows.h>

// Shared variable
int counter = 0;

// Mutex handle
HANDLE mutex;

// Function executed by threads
DWORD WINAPI threadFunction(LPVOID arg)
{
    int i;

    for (i = 0; i < 1000000; i++)
    {
        // Lock the mutex
        WaitForSingleObject(mutex, INFINITE);

        // Critical section
        counter++;

        // Unlock the mutex
        ReleaseMutex(mutex);
    }

    return 0;
}

int main()
{
    HANDLE thread1, thread2;

    // Create mutex
    mutex = CreateMutex(NULL, FALSE, NULL);

    if (mutex == NULL)
    {
        printf("Failed to create mutex.\n");
        return 1;
    }
}
```

```

// Create threads
thread1 = CreateThread(
    NULL,
    0,
    threadFunction,
    NULL,
    0,
    NULL
);

thread2 = CreateThread(
    NULL,
    0,
    threadFunction,
    NULL,
    0,
    NULL
);

// Wait for both threads to finish
WaitForSingleObject(thread1, INFINITE);
WaitForSingleObject(thread2, INFINITE);

// Cleanup
CloseHandle(thread1);
CloseHandle(thread2);
CloseHandle(mutex);

// Print result
printf("Final Counter Value: %d\n", counter);

return 0;
}

```

## OUTPUT:

```

● PS C:\Users\SMILE\Documents\OS\Lab programs> ./lab_19.exe
Final Counter Value: 2000000

```